AI-Driven Automation of Construction Cost Estimation: Integrating BIM with Large Language Models

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Sebastian Herrera

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AUTHORS → Mohamed Abdelsalam, Amr Ashmawi and Phuong H. D. Nguyen

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Mohamed Abdelsalam [1], Amr Ashmawi [2] and Phuong H. D. Nguyen [3,*]

MAIN FINDINGS AND ARGUMENTS

- The study proposes an AI-driven framework that integrates BIM, Large Language Models (LLMs), and the Model Context Protocol (MCP) for automated construction cost estimation.
- The system automatically extracts quantities from BIM models and connects them with industry cost databases.
- AI performs component classification, cost matching, and cost calculations with minimal human input.
- In the case study, the automated workflow reduced estimation time from 2.5–3.5 hours to about 42 seconds.
- The automated estimate achieved a cost variance of approximately 5.1% compared to manual estimates, showing competitive accuracy.

BROADER IMPLICATIONS FOR SOCIETY

- AI automation can significantly improve productivity and efficiency in the construction industry.
- Automated workflows reduce human errors and inconsistencies in cost estimation.
- Faster and more accurate estimates can improve project planning and financial decision-making.
- Conversational AI interfaces allow professionals to interact with complex systems using natural language instead of programming skills.
- Increased automation may contribute to lower project costs, faster delivery times, and improved infrastructure planning.

WHY IT MATTER IN CONSTRUCTION?

- The framework demonstrates how AI and BIM integration can automate key construction management tasks.
- Standardized protocols like MCP allow easier integration between AI systems, BIM software, and cost databases.
- The approach could significantly reduce the time required for cost estimation workflows.
- The system provides a scalable model for future automation in construction management processes.
- The research offers a practical blueprint for adopting AI technologies in the AEC (Architecture, Engineering, and Construction) industry.